

## DALLAS LOVE FIELD, TX, USA

WMO: 722580

Lat: 32.852N

Lon: 96.856W

Elev: 134

StdP: 99.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13960

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.1       | -2.0 | -13.2                         | 1.2 | 0.2  | -10.8 | 1.5 | 2.0  | 11.5                  | 8.4  | 10.6 | 9.4  | 4.4                   | 350  | 0.515 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>10.0            | (c)<br>38.7     | (d)<br>23.7 | (e)<br>37.5 | (f)<br>24.0 | (g)<br>36.3 | (h)<br>23.9 | (i)<br>26.2         | (j)<br>33.7 | (k)<br>25.6 | (l)<br>33.1 | (m)<br>25.2 | (n)<br>32.5 | (o)<br>4.0           | (p)<br>170 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.1                            | 19.3 | 29.0 | 23.7 | 18.8 | 28.7 | 23.0 | 18.1 | 28.3 | 81.7          | 33.8 | 79.0 | 32.9 | 77.1 | 32.5 | 33.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 10.5          | (b) 9.2 | (c) 8.4 | (e) -7.4                   | (f) 40.6 | (g) 1.8            | (h) 1.6  | (i) -8.7                                           | (j) 41.8  | (k) -9.7   | (l) 42.7  | (m) -10.7  | (n) 43.6  | (o) -12.0  | (p) 44.8  |    |    |
| (4) 10.5          | (5) 9.2 | (6) 8.4 | (7) -7.4                   | (8) 40.6 | (9) 1.8            | (10) 1.6 | (11) -8.7                                          | (12) 41.8 | (13) -9.7  | (14) 42.7 | (15) -10.7 | (16) 43.6 | (17) -12.0 | (18) 44.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 20.0   | 8.8   | 11.0  | 15.2  | 19.5  | 24.1  | 28.4  | 30.5  | 30.7  | 26.7  | 20.7  | 14.5  | 9.6  |
|                                                                                  | DBStd       | 8.91   | 5.60  | 5.93  | 5.26  | 4.05  | 3.51  | 2.39  | 2.30  | 2.57  | 3.58  | 4.50  | 5.02  | 5.28 |
|                                                                                  | HDD10.0     | 249    | 89    | 54    | 17    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 16    | 71   |
|                                                                                  | HDD18.3     | 1109   | 298   | 212   | 123   | 34    | 4     | 0     | 0     | 0     | 1     | 29    | 133   | 274  |
|                                                                                  | CDD10.0     | 3908   | 53    | 82    | 177   | 287   | 438   | 552   | 637   | 641   | 502   | 331   | 151   | 59   |
|                                                                                  | CDD18.3     | 1724   | 2     | 7     | 25    | 69    | 183   | 302   | 378   | 382   | 253   | 101   | 17    | 4    |
|                                                                                  | CDH23.3     | 20782  | 13    | 47    | 168   | 523   | 1754  | 3719  | 5282  | 5368  | 2918  | 868   | 103   | 18   |
|                                                                                  | CDH26.7     | 10622  | 1     | 8     | 28    | 120   | 670   | 1874  | 3049  | 3149  | 1420  | 287   | 14    | 1    |
| Wind                                                                             | WSAvg       | 4.3    | 4.1   | 4.4   | 4.9   | 5.1   | 4.9   | 4.6   | 4.1   | 3.7   | 3.5   | 3.9   | 4.0   | 3.9  |
| Precipitation                                                                    | PrecAvg     | 927    | 49    | 69    | 77    | 95    | 120   | 95    | 58    | 57    | 83    | 104   | 61    | 64   |
|                                                                                  | PrecMax     | 1405   | 103   | 201   | 231   | 391   | 268   | 276   | 216   | 152   | 271   | 408   | 178   | 235  |
|                                                                                  | PrecMin     | 445    | 0     | 6     | 7     | 1     | 14    | 8     | 0     | 0     | 1     | 0     | 4     | 1    |
|                                                                                  | PrecStd     | 233    | 31    | 44    | 52    | 71    | 69    | 68    | 52    | 43    | 61    | 87    | 42    | 56   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.2  | 27.6  | 29.4  | 31.8  | 35.4  | 37.8  | 40.1  | 40.8  | 38.5  | 33.6  | 28.7  | 25.8 |
|                                                                                  |             | MCWB   | 15.9  | 16.0  | 18.7  | 21.0  | 23.3  | 24.1  | 23.7  | 23.3  | 22.8  | 22.1  | 19.6  | 17.6 |
|                                                                                  | 2%          | DB     | 22.5  | 24.6  | 27.1  | 29.5  | 33.4  | 36.4  | 38.5  | 39.0  | 36.3  | 31.7  | 26.1  | 22.7 |
|                                                                                  |             | MCWB   | 15.2  | 16.5  | 18.3  | 20.4  | 23.1  | 24.0  | 23.7  | 23.8  | 23.3  | 21.7  | 18.9  | 16.4 |
|                                                                                  | 5%          | DB     | 20.3  | 22.3  | 25.2  | 27.9  | 31.8  | 35.1  | 37.3  | 37.7  | 34.7  | 29.8  | 24.1  | 20.7 |
|                                                                                  |             | MCWB   | 14.8  | 16.0  | 17.2  | 19.6  | 22.6  | 23.9  | 24.1  | 24.0  | 23.0  | 20.6  | 17.9  | 15.6 |
|                                                                                  | 10%         | DB     | 18.0  | 20.3  | 23.3  | 26.4  | 30.3  | 33.9  | 36.1  | 36.5  | 33.1  | 28.0  | 22.5  | 18.5 |
|                                                                                  |             | MCWB   | 13.3  | 15.3  | 16.7  | 18.9  | 22.1  | 23.8  | 24.1  | 23.9  | 22.7  | 19.9  | 17.2  | 14.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 19.1  | 20.2  | 21.2  | 23.5  | 25.8  | 27.1  | 27.0  | 26.9  | 25.7  | 24.5  | 22.0  | 19.9 |
|                                                                                  |             | MCDB   | 22.0  | 23.2  | 25.9  | 28.5  | 31.4  | 33.9  | 35.4  | 34.7  | 33.3  | 29.1  | 25.9  | 22.8 |
|                                                                                  | 2%          | WB     | 17.9  | 19.1  | 20.1  | 22.2  | 24.8  | 25.9  | 26.0  | 25.8  | 25.0  | 23.6  | 20.6  | 18.5 |
|                                                                                  |             | MCDB   | 20.4  | 22.0  | 24.4  | 27.1  | 30.4  | 32.8  | 34.0  | 33.7  | 32.1  | 28.4  | 23.8  | 21.7 |
|                                                                                  | 5%          | WB     | 16.3  | 17.8  | 19.2  | 21.4  | 24.0  | 25.2  | 25.5  | 25.3  | 24.5  | 22.8  | 19.4  | 16.8 |
|                                                                                  |             | MCDB   | 19.3  | 21.0  | 23.4  | 25.8  | 29.5  | 31.9  | 33.3  | 33.3  | 30.9  | 27.5  | 22.9  | 19.7 |
|                                                                                  | 10%         | WB     | 13.8  | 15.9  | 18.0  | 20.5  | 23.2  | 24.7  | 25.0  | 24.9  | 24.0  | 21.8  | 18.1  | 14.8 |
|                                                                                  |             | MCDB   | 17.2  | 19.7  | 22.3  | 24.7  | 28.8  | 31.2  | 32.6  | 32.8  | 30.0  | 26.4  | 21.8  | 17.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.4  | 10.4  | 10.6  | 10.6  | 9.8   | 9.6   | 9.9   | 10.0  | 10.1  | 10.6  | 10.3  | 9.9  |
|                                                                                  |             | MCDBR  | 13.6  | 13.0  | 12.4  | 11.7  | 10.9  | 10.6  | 11.0  | 11.2  | 11.3  | 11.9  | 11.7  | 12.7 |
|                                                                                  |             | MCWBR  | 8.3   | 7.9   | 6.5   | 5.8   | 4.1   | 3.0   | 2.8   | 2.7   | 3.3   | 4.9   | 6.6   | 7.9  |
|                                                                                  | 5% WB       | MCDBR  | 10.7  | 10.8  | 10.4  | 9.6   | 9.6   | 9.6   | 9.9   | 10.0  | 9.4   | 9.5   | 10.0  | 10.5 |
|                                                                                  |             | MCWBR  | 8.4   | 8.2   | 7.0   | 6.1   | 4.7   | 3.4   | 3.1   | 2.9   | 3.4   | 4.8   | 7.6   | 8.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.317  | 0.331 | 0.361 | 0.399 | 0.430 | 0.443 | 0.445 | 0.447 | 0.404 | 0.361 | 0.340 | 0.322 |      |
|                                                                                  | taud        | 2.496  | 2.467 | 2.399 | 2.314 | 2.288 | 2.285 | 2.318 | 2.296 | 2.366 | 2.471 | 2.470 | 2.486 |      |
|                                                                                  | Ebn at Noon | 903    | 924   | 916   | 891   | 861   | 846   | 842   | 837   | 861   | 878   | 866   | 875   |      |
|                                                                                  | Edh at Noon | 86     | 98    | 113   | 129   | 134   | 134   | 129   | 130   | 116   | 97    | 88    | 83    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.86   | 3.34  | 4.43  | 5.39  | 5.81  | 6.54  | 6.68  | 6.13  | 5.15  | 4.04  | 3.04  | 2.52  |      |
|                                                                                  | RadStd      | 0.28   | 0.52  | 0.35  | 0.41  | 0.53  | 0.62  | 0.45  | 0.40  | 0.45  | 0.60  | 0.39  | 0.32  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (47 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +106    | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page